Spirotrichonymphea

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The Spirotrichonymphea are a group of obligate symbionts of termites (Taerum et al., 2020). They belong to the lineage Parabasalia within the protist clade Metamonada (Archibald et al., 2017). Similarly to other members of the Metamonada group, Spirotrichonymphea are anaerobic, flagellated heterotrophs. A notable feature that sets these organisms apart from most other metamonads is the number of flagella they possess. Spirotrichonymphea are hypermastigotes and thus belong to a polyphyletic group of multiflagellates. Their flagella are arranged in two or more characteristic rows that coil around the cell (Gile et al., 2018). Initially, the great number of flagella was thought to mean that these organisms are ciliates, however, they are currently described as one of the six classes of Parabasalia (Archibald et al., 2017). Spirotrichonymphea are well adapted to survive in the hindguts of their wood-eating hosts. This symbiotic relationship is so well-established that it is believed it emerged 150 million years ago, before the divergence of the termite and wood roach Cryptocercus lineages (Taerum et al., 2020). The gut-inhabiting protists are transmitted to new insect hosts via parent-to-offspring proctodeal trophallaxis, a form of social behavior (Nalepa, 2015). It involves the recipient consuming the donor's hindgut fluids. Thus, Spirotrichonymphea are passed from one insect generation to the next: they have become highly specialized and even co-evolved with their hosts (Archibald et al., 2017). The dependence is mutual, as the host might face starvation if its gut symbionts are removed. Spirotrichonymphea play a significant role in the breakdown of lignocellulose, an abundant yet hard-to-digest food source and a constituent of wood (Taerum et al., 2020). However, they do not act independently in the digestion process: Spirotrichonymphea themselves are highly reliant on bacterial symbionts (Archibald et al., 2017). Bacteria can sometimes be observed adhering to the cell surface, see Figure 1 (Taerum et al., 2020). The symbiotic bacteria might benefit their hosts by fixing nitrogen and producing acetate (Ohkuma et al., 2015). They might even aid Spirotrichonymphea's motility (Archibald et al., 2017). Despite the observed diversity and inferred ecological importance of Spirotrichonymphea, they are poorly studied (Taerum et al., 2020). There are several difficulties associated with culturing these organisms, as the conditions they require are highly specific (Archibald et al., 2017). Hopefully, future research will uncover more insights into Spirotrichonymphea and the complex relationships between these protists and their symbionts.

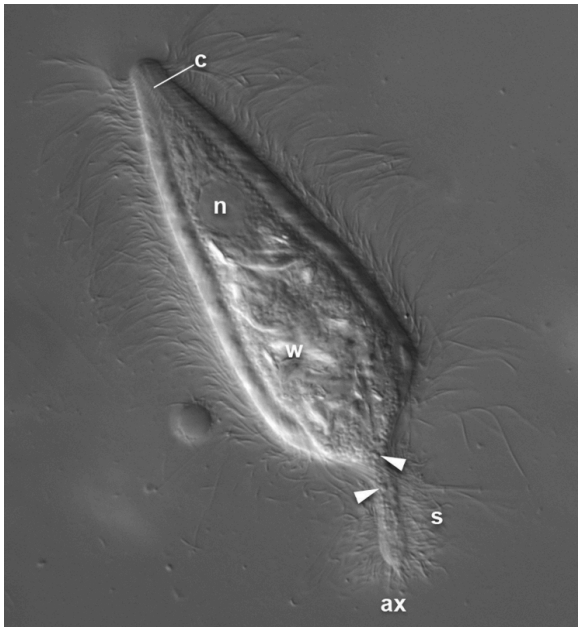


Figure 1 adapted from Taerum et al. (2020): a microscopy image of *Cuppa polygyra*, a species of Spirotrichonymphea. Spirochete bacteria (s) are attached to the cell surface.

References

Archibald, J. M., Simpson, A. G. B., & Slamovits, C. H. (2017). Handbook of the protists. Springer.

Gile, G. H., James, E. R., Tai, V., Harper, J. T., Merrell, T. L., Boscaro, V., Husník, F., Scheffrahn, R. H., & Keeling, P. J. (2018). New Species of Spirotrichonympha from Reticulitermes and the Relationships Among Genera in Spirotrichonymphea (Parabasalia). The Journal of Eukaryotic Microbiology, 65(2), 159–169. https://doi.org/10.1111/jeu.12447

Nalepa, C. A. (2015). Origin of termite eusociality: trophallaxis integrates the social, nutritional, and microbial environments. Ecological Entomology, 40(4), 323–335. https://doi.org/10.1111/een.12197

Ohkuma, M., Noda, S., Hattori, S., Iida, T., Yuki, M., Starns, D., Inoue, J., Darby, A. C., & Hongoh, Y. (2015). Acetogenesis from H2plus CO2and nitrogen fixation by an endosymbiotic spirochete of a termite-gut cellulolytic protist. Proceedings of the National Academy of Sciences, 112(33), 10224–10230. https://doi.org/10.1073/pnas.1423979112

Taerum, S. J., Jasso-Selles, D. E., Hileman, J. T., De Martini, F., Mizumoto, N., & Gile, G. H. (2020). Spirotrichonymphea (Parabasalia) symbionts of the termite Paraneotermes simplicicornis. European Journal of Protistology, 76, 125742. https://doi.org/10.1016/j.ejop.2020.125742